The documentation and process conversion measures necessary to comply with this revision shall be completed by 26 October 1999.

INCH-POUND

MIL-PRF-19500/567A
26 July 1999
SUPERSEDING
MIL-S-19500/567
13 April 1987

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON,
SCHOTTKY BARRIER, FAST RECOVERY
TYPE 1N6492, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, fast recovery, Schottky barrier, semiconductor diode. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500 (see 6.3.2).

1.2 Physical dimensions. See figure 1 (similar to T0-205AF).

1.3 Maximum ratings.

Type	V_{RRM} and V_{RWM} 1/	V_{RSM}	V_R 1/	I_F 1 (AV) 2/ T_A = 25°C 4/	I_F 1 (AV) 2/ T_C = 100°C	I_O T_C = 100°C 3/	I_{FSM}	T_{op} and T_{STG}
	V (pk)	V (pk)	V dc	A dc	A dc	A dc	A (pk)	°C
1N6492	45	54	45	1.20	4	3.60	80	-65 to +175

1/ Full rated V_{RRM} and V_{RWN} with appropriate average forward current (see footnote 2/) is applicable over the range of T_C from -55°C to +135°C. Full rated V_R is applicable over the range of T_C from -55°C to +120°C. With these maximum voltages and case temperatures, $T_J \leq 175$°C.

2/ Average current with a 50 percent duty cycle square wave including reverse voltage amplitude equal to the magnitude of full rated V_{RWM}. Derate linearly at 114 mA dc/°C for $T_C > 100$°C (to 0 at T_C = 135°C); if V_{RWM} = 20, derate I_F (AV) at 62 mA/°C, to 0 at T_C = 165°C.

3/ Average current with an applied sine wave including reverse voltage equal to the magnitude of full rated V_{RWM}. Derate linearly at 103 mA dc/°C for $T_C > 100$°C ; if V_{RWN} = 20, derate at 55 mA/°C.

4/ This rating requires no special mounting, heat sinking, or forced air flow across the device.

1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

TYPE	V_{FM1} I_{FM} = 4 A (pk)	V_{FM2} I_{FM} = 2 A (pk)	I_{RM} V_{RM} = 45 V (pk) T_J = 125°C	I_{RM} V_{RM} = 45 V (pk) T_J = 25°C	C_T V_R = 5 V dc	$R_{\theta JC}$	$R_{\theta JA}$
	V (pk)	V (pk)	mA (pk)	mA (pk)	pF	°C/W	°C/W
1N6492	0.68	0.56	20	2.0	450	12.0	175

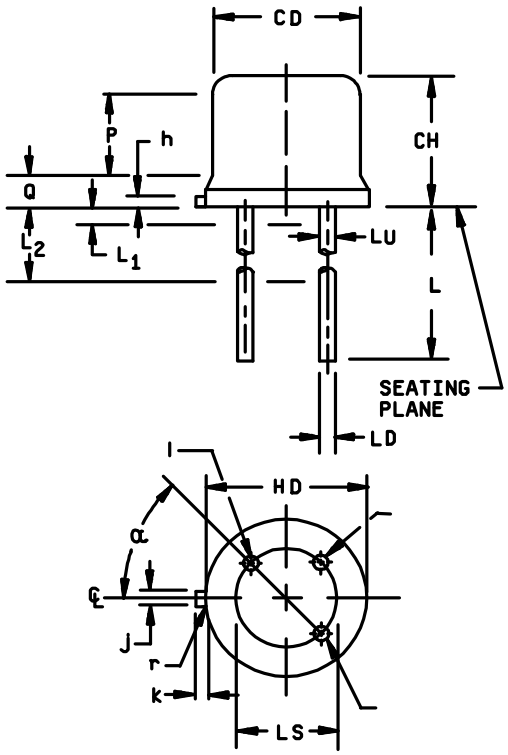
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.160	.180	4.07	4.57	
HD	.335	.370	8.51	9.40	
LD	.016	.021	0.41	0.53	8, 9
L	.500	.750	12.7	19.05	8, 9
LS	.200 TP		5.08 TP		7
LU	.016	.019	0.41	0.48	8, 9
L1		.050		1.27	8, 9
L2	.250		6.35		8, 9
P	.100		2.54		6
Q		.040		1.02	5
r		.010		0.018	10
TL	.029	.045	0.74	1.14	
TW	.028	.034	0.72	0.86	
α	45° TP		45° TP		7
Term 1	Anode				
Term 2	Open (no connection)				
Term 3	Cathode (case)				



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond radius (r) maximum, j shall be held for a minimum length of .011 inch (0.028 mm).
4. Dimension k measured from maximum HD.
5. Outline in this zone is not controlled.
6. Dimension CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
8. LU applies between L_1 and L_2. LD applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
9. All three leads.
10. Radius (r) applies to both inside corners of tab.
11. Cathode is electrically connected to the case.
12, In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-205AF).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein.

3.4 Marking. Devices shall be marked as specified in MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	Test condition F t_{LOW} = -55°C and T_{HIGH} = +175°C	Test condition F t_{LOW} = -55°C and T_{HIGH} = +175°C
9	Reverse energy test (see 4.5.2) V_{FM2} , I_{R1}	Not applicable
10	T_C = 130°C	T_C = 130°C
11	Reverse energy test (see 4.5.2) ΔV_{FM2} ± 50 mV of initial value ΔI_{R1} ± 100 % or 500 µA, whichever is greater. Subgroup 2 of table I herein.	Reverse energy test (see 4.5.2) V_{FM2} , I_{R1} Subgroup 2 of table I herein.
12	See 4.3.1	See 4.3.1
13	Reverse energy test (see 4.5.2) ΔV_{FM2} ± 50 mV ΔI_{R1} ± 100 % or 500 µA, whichever is greater. Scope display evaluation (see 4.5.5)	Reverse energy test (see 4.5.2) ΔV_{FM2} ± 50 mV ΔI_{R1} ± 100 % or 500 µA, whichever is greater. Scope display evaluation (see 4.5.5)

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows (utilizing MIL-STD-750, method 1038, condition B):

T_A = Room ambient as defined in 4.5 of MIL-STD-750.
V_R = 45 V (pk)
I_O = 0.75 A dc
f = 60 Hz

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and paragraphs 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1056	Thermal shock, condition F, T_{LOW} = -55°C, T_{HIGH} = +175°C
B4	1037	I_O = 0.75 A, T_A = 25°C; V_{RM} = 45 V, t_{on} = t_{off} = 3 minutes for a minimum of 2,000 cycles.
B5	1027	I_F = 3.2 A dc or adjust T_A or I_F as required by the corresponding I_F or T_A to achieve a lot T_J = 275°C
B6	4081	See 4.5.3 $R_{\theta JC}$ ≤ 12.0°C/W

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
B2	1056	Thermal shock, condition F, T_{LOW} = -55°C, T_{HIGH} = +175°C
B3	1027	T_A = 25°C, I_O = 0.75 A, V_{RN} = 45 V (pk)
B3	2037	Condition A, all internal wires must be pulled separately.
B5	4081	See 4.5.3. $R_{\theta JC} \leq$ 12.0°C/W
B6	1032	T_A = +175°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C6	1026	I_O = 0.75 A, V_{RWN} = 45 V(pk), T_A = 25°C
C7	----	Operational Power cycling, see 4.5.6, $T_{C(LOW)}$ = 40°C, +0, -15°C; $T_{C(HIGH)}$ = +115°C +5, -0°C, 5000 cycles, n = 22, c = 0,

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 2 of MIL-STD-750.

4.5.2 Peak reverse energy test. The peak reverse energy test is to be performed as shown on figure 2 or equivalent. The diode under test must be capable of absorbing the reverse energy, as defined, and meet the electrical requirements of table I, subgroup 4 herein.

4.5.3 Thermal resistance. Thermal resistance shall be measured in accordance with MIL-STD-750, method 4081. The case reference temperature shall be held to equilibrium within the range of +20°C to +70°C during the power application, and shall be measured at the hottest part of the case. The following measurements shall apply:

I_{F1} = 1 A at 97 percent minimum duty factor
I_{F2} = 10 mA dc

4.5.4 Reverse current at peak reverse voltage, alternate test. The reverse current at peak reverse voltage test may be satisfied by performing the reverse energy test of 4.5.2 and measuring breakdown voltage to ensure $V_{(BR)} \geq$ 54 V (pk) with I_{RM} = 2.0 A (pk)

4.5.5 Scope display evaluation. Scope display evaluation shall be in accordance MIL-STD-750, method 4023.

4.5.6 Operational power cycling (qualification only). One complete cycle for this test shall consist of the following two steps: Step 1, heat the case to the $T_{c(high)}$ specified, by passing forward current through the diode under test. The reverse voltage shall be only enough to permit the reverse current to flow, and should be a maximum of 5 volts. Step 2, remove the applied current allow the case temperature to cool to the $T_{C(LOW)}$ specified. Forward current shall be chosen to achieve the $T_{c(high)}$ condition in 75 $\pm$50 seconds. The cycling must be continuous until the required number of cycles has been completed. It is permissible to force cool the device during step 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection

1/ Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2		2/				
Forward voltage	4011	I_{FM} = 4.0 A(pk), pulsed (see 4.5.1)	V_{FM2}		0.68	V
	4011	I_{FM} = 2.0 A(pk), pulsed (see 4.5.1)	V_{FM3}		0.56	V
	4011	I_{FM} = 1.0 A(pk), pulsed (see 4.5.1)	V_{FM4}		0.48	V
	4011	I_{FM} = 8.0 A(pk), pulsed (see 4.5.1)	V_{FM1}		0.92	V
Reverse current leakage	4016	V_{RM} = 45 V(pk), pulsed method, (see 4.5.1)	I_{RM1}		2.0	mA
Subgroup 3		2/				
High temperature operation		T_A = +125°C,				
Reverse current leakage	4016	V_{RM} = 45 V(pk), pulsed method, (see 4.5.1)	I_{RM2}		20	mA
High temperature operation		T_A = +175°C				
Reverse current leakage	4016	V_{RM} = 45 V(pk), pulsed method, (see 4.5.1)	I_{RM3}		200	mA
Low temperature operation		T_A = -55°C				
Reverse current leakage	4016	V_{RM} = 45 V(pk), pulsed method, (see 4.5.1)	I_{RM3}		20	mA
Forward voltage	4011	I_{FM} = 2.0 A(pk), pulsed (see 4.5.1)	V_{FM3}		0.63	V
Subgroup 4						
Reverse current leakage at peak reverse voltage	4016	Pulsed method (see 4.5.1) V_{RSM} = 54 V(pk), (alternate test, see 4.5.4) 2/	I_{RM4}		2.0	A
Capacitance	4001	V_R = 5 V, .01 ≤ f ≤ 1 MHz V_{SIG} = 15 mV(p-p)	C_T		450	pF
Subgroup 5						
Surge current	4066	I_{FSM} = 80 A(pk), V_{RM} = 45 V(pk), I_O = 0.75 A 10 surges of 8.3 ms each at 1 minute intervals, T_A = 25°C				
Electrical measurements		See table I, group A, subgroup 2 herein				

1/ See MIL-PRF-19500 for sampling plan.
2/ Measurement point is on the lead 0.2 inches below the seating plane of the case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

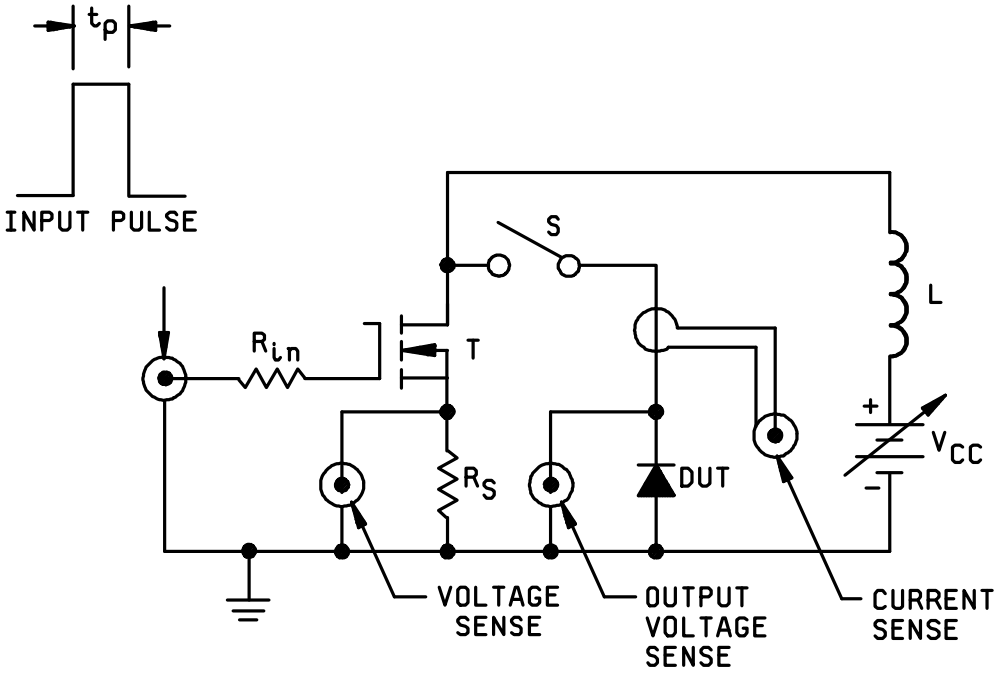
TABLE II. Groups B and C delta tests. 1/, 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward Voltage	4011	I_{FM} = 4 A(pk) pulsed (see 4.5.1)	ΔV_{FM2}	±50 mV change from previous value		V
2	Reverse current	4016	V_{RM} = 45 V(pk) pulsed (see 4.5.1)	ΔI_{FM1}	+500µA dc or +100% of initial value, whichever is greater		mV

1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
Subgroups 4 and 5, see table II herein, steps 1 and 2.

2/ The delta measurements for table VII of MIL-PRF-19500 are as follows:
Subgroup 6, see table II herein, steps 1 and 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.



V_G = 10 Volts	R_{in} = 5.0 Ωohms, 1 W
R_G = 50 Ω	R_S = 0-1 Ω, 1 W
PW = 30 µs	L = 260 µH
Duty cycle ≤ 1 %	T = IRF130/2N6756 or equivalent

Procedure:

1. With S open, adjust pulse width to test current of 2 A across R_S .
2. Close S. Verify test current with current sense.
3. Read peak output voltage (see 4.5.4)

FIGURE 2. Peak reverse energy test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personal need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agencys' automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES:

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Product assurance level and type designation.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-1992)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/567A	2. DOCUMENT DATE 99/07/26

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, TYPES JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY				
a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAC Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC -LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvior, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

DD FORM 1426, FEB 99 (EG) PREVIOUS EDITIONS ARE OBSOLETE WHS/DIOR, Feb 99

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:29Z
Check the source to verify that this is the current version before use.